COMMUNITY VI

Clinical Postings | 3rd Year MBBS

Hi Sammy!
Field work = real-world medicine! Go to the community & learn!

Community Visits | Clinical Postings | 3rd Year | MBBS

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Family Health Survey Methodology

Family survey = your
first field experience!
Follow the steps!

FAMILY HEALTH SURVEY

A systematic door-to-door survey to assess health status, morbidity & determinants of a family/community.

Steps of Conducting a Family Health Survey

1	Planning	Define objectives, select area, prepare tools, team formation
2	Sampling	Random/systematic sampling of households from voter list or census
3	Tool Prep	Design proforma/questionnaire, pre-test on pilot families
4	Field Visit	Door-to-door visit, introduce, consent, interview, examine
5	Data Entry	Record data systematically, check for completeness
6	Analysis	Calculate rates, ratios, proportions, identify health problems
7	Report	Community diagnosis report with recommendations
8	Feedback	Health education session + camp for the community

Data Collected in Family Survey

Section	Details
Demographic	Family size, age-sex distribution, religion, caste, SES
Socioeconomic	Occupation, income, BG Prasad / Kuppuswamy scale
Environmental	Housing type, water source, sanitation, waste disposal
Maternal/Child	ANC visits, immunization status, nutrition, birth weight
Morbidity	Current illnesses, chronic diseases, disabilities
Health Behavior	Smoking, alcohol, diet, exercise, treatment-seeking
Health Services	Utilization of SC/PHC, ASHA contact, unmet needs

Community diagnosis =
diagnosing a VILLAGE!
Just like a patient!

Community Diagnosis — Concept

- Definition: Identifying health problems & determinants of a community
- Analogous to clinical diagnosis but for a POPULATION, not individual
- Steps: Data collection -> Analysis -> Prioritization -> Action plan
- Uses: Identify disease burden, plan interventions, allocate resources
- Done during: Community Medicine posting (rural/urban field area)

Health Survey Tools + Questionnaires

Tool Type	Details & Examples
Structured Q	Pre-coded, fixed options (e.g., Yes/No, MCQ) — easy to analyze
Semi-structured	Mix of open + closed questions — flexible yet standardized
Unstructured	Open-ended, in-depth — qualitative research, FGD, interview
Rating Scales	Likert scale (1-5), Visual Analog Scale (VAS) for pain/severity
Validated Tools	PHQ-9 (depression), WHO QoL, SES scales (BG Prasad/Kuppuswamy)
Observation	Checklist-based observation of sanitation, hygiene, environment

Socioeconomic Status (SES) Scales

Scale	Setting	Parameters Used
Kuppuswamy	Urban	Education + Occupation + Income (3 factors)
BG Prasad	Urban/Rural	Per capita monthly income (updated with CPI)
Pareek	Rural	Caste, occupation, education, land, housing + more
BPL (Below PL)	Government	Scoring system — 0-4 for each indicator

★ Kuppuswamy = EOI!
Education + Occupation + Income = 3 parameters for urban SES classification
BG Prasad uses ONLY income (simpler!). Both updated with CPI regularly.

Epidemiological Study + Sample Size

Sample size & study design = always asked! Know the formulas!

Types of Epidemiological Studies (Field Work)

Study Type	Design	Use in Field
Cross-sectional	One time point, prevalence	Family health survey, KAP study
Case-control	Cases vs controls, retrospective	Outbreak investigation
Cohort	Follow-up, prospective/retro	Long-term disease risk assessment
Experimental	Intervention + control group	Vaccine trial, health education RCT
Descriptive	Who, Where, When	Disease pattern description, surveillance

Sample Size Determination

- For prevalence study: n = Z^2 x P(1-P) / d^2
- Z = 1.96 (for 95% CI), P = expected prevalence, d = precision
- For comparing means: n = 2(Z_alpha + Z_beta)^2 x SD^2 / (mean diff)^2
- Rule of thumb: Larger sample = more precise, but costlier
- Non-response: Add 10-20% extra to calculated sample size

Data Collection Methods in Field

Method	Details
Interview	Face-to-face, structured/semi-structured questionnaire
Observation	Direct observation with checklist (sanitation, hygiene)
Record Review	Existing records: birth/death register, immunization cards
Focus Group	FGD: 6-12 participants, trained moderator, qualitative data
Key Informant	Interview village leader, ASHA, AWW for community insights
Anthropometry	Height, weight, MUAC, BMI — for nutrition assessment

Sampling Methods — Quick Reference

Method	How It Works
Simple Random	Every unit has equal chance (lottery, random number table)
Systematic	Every kth unit (e.g., every 5th house on a street)
Stratified	Divide into strata (age, sex), then random from each
Cluster	Divide into clusters (villages), randomly select clusters
Multistage	Combine methods: district -> block -> village -> household

Field work questions are scoring! Tables & steps = full marks!

Most Repeated LAQs (10–12 marks)

- 1. Describe the steps of conducting a family health survey in a community
- 2. What is community diagnosis? Describe methodology & report format
- 3. Types of epidemiological studies — classify with examples
- 4. Describe sampling methods used in community surveys

Most Repeated SAQs (5–6 marks)

- 1. Sample size formula for prevalence study — explain each component
- 2. Kuppuswamy scale — parameters, classes, limitations
- 3. Difference between structured vs unstructured questionnaire
- 4. Steps of community diagnosis with examples
- 5. Types of sampling methods with advantages/disadvantages
- 6. Data collection methods used in field surveys

One-Liner Associations (MCQ/Viva Gold)

Question	Answer
Family health survey is which study type?	Cross-sectional (descriptive)
Kuppuswamy scale parameters	Education + Occupation + Income
BG Prasad uses which single parameter?	Per capita monthly income
Z value for 95% confidence interval	1.96
Cluster sampling advantage	Practical, cheaper for large areas
30x7 cluster sampling used for	Immunization coverage survey (WHO)
Qualitative data collection method	Focus Group Discussion (FGD)
Best sampling for rare disease study	Case-control (retrospective)
Non-response rate correction	Add 10-20% to sample size
Community diagnosis is analogous to	Clinical diagnosis for a population

♥ Going to the field teaches you what textbooks can't, Sammy!
The community is your patient. Learn to listen to it!